Practice workbook · High-level and low-level languages

Characteristics of high- and low-level languages, their benefits and drawbacks, and why a task suits one or the other.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-15

1. Question 5 · part b.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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1 .....

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2 .....

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(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

1. Question 5 · part b.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

(2)

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

(2)

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

(1)

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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P 7 8 6 3 7 A 0 1 4 2 0

1. Question 5 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 6 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

6 A train company uses ticket vending machines at each station.

(a) The machines use embedded systems.

(i) Explain **one** benefit of using an embedded system in these machines.

(2)

(ii) Customers use a touch screen to select their destination. They can pay by cash or bank card. Their tickets and a receipt are printed.

The touch screen is controlled by an embedded system.

Give **two** other hardware components in the ticket machine that are controlled by embedded systems.

(2)

1 .....

2 .....

(iii) The ticket machine uses data encryption when a customer pays using a bank card.

State why data encryption is used in this case.

(1)

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P 6 1 8 8 0 A 0 2 0 2 4

2. Question 6 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

(b) Compare **four** features of high-level and low-level programming languages.

(4)

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2. Question 6 · part b · 4 marks

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Further practice · Attempt part b only.

(c) Artificial intelligence (AI), in many forms, has an increasing impact on our lives.

Discuss this statement considering characteristics, uses and ethical issues.

(6)

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(Total for Question 6 = 15 marks)

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2. Question 6 · part b · 4 marks · Working and answer

Working and answer

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3. Question 6 · part a.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a.i only.

6 Akiko develops software programs.

(a) Akiko can use a low-level or a high-level programming language to write code.

(i) Give **one** benefit of using a low-level programming language to develop a program.

(1)

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.....

(ii) Give **one** drawback of using a low-level programming language to develop a program.

(1)

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(b) Akiko is writing an anti-virus program.

(i) The program will include a virus removal function.

Give **two other** functions that should be included.

(2)

1 .....

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2 .....

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(ii) An anti-virus program is one type of anti-malware.

Give **one other** type of anti-malware.

(1)

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3. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

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4. Question 6 · part a.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part a.ii only.

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4. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part e.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part e.i only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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5. Question 1 · part e.i · 1 mark

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Further practice · Attempt part e.i only.

(d) Identify which **one** of these describes a sequential computational model.

(1)

- ☒ **A** Program instructions are read one after another from external storage
- ☒ **B** Program instructions are executed by multiple agents working together
- ☒ **C** Program instructions are executed in parallel by different cores
- ☒ **D** Program instructions are executed one after another

(e) A program can be written in a high-level or a low-level language.

(i) Give **one** reason for writing a program in a low-level language.

(1)

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(ii) State the purpose of an assembler.

(1)

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(iii) Complete the table by adding **one** tick (✓) in **each** row to match the description.

(3)

Description	Compiler	Interpreter
Translates the program each time it is executed		
Produces permanent object code		
Translates line by line		
Translates the whole program before it is run		
Generates a list of errors once the complete program has been translated		

(Total for Question 1 = 11 marks)

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5. Question 1 · part e.i · 1 mark · Working and answer

Working and answer

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